



CPaaS User Segmentation Analysis:

Personalization, Retention & Upsell Opportunities

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Executive Summary

Project Description : Segmented 20 enterprise customers using an RFM-based framework adapted for CPaaS.

Objective : Identify High-value clients, Churn-risk clients, and Upsell opportunities

Tools : Python

Result : 3 behavioral segments: Champions, Loyal Clients, and Low-Intensity

Impact : These segments enable targeted retention, infrastructure optimization, and revenue expansion strategies.

Business Background

OCA Indonesia is a Communication Platform-as-a-Service (CPaaS) owned by Telkom Indonesia. The company mission to empower businesses to communicate at scale via WhatsApp, SMS, Email, and Voice Call, seamlessly integrated through dashboard sand API integrations.

OCA manages three distinct client traffic patterns: Single Blasts, Scheduled Blast, and Repeated Blasts.

To stay a trusted CPaaS leader, OCA needs to manage high-volume traffic efficiently and personalize engagement by client profile.

Business Problem

- All users are treated similarly in support and pricing, regardless of their infrastructure impact or revenue contribution.
- OCA needs a behavioral architecture to drive personalization, mitigate backend churn, and run high-margin cross-sell campaigns.

Root Cause and Objective

Across all 20 users, our Voice Call channel suffers from a universal ~78% failure rate, while WhatsApp and SMS remain highly stable.

Build a simple **RFM-based segmentation** of CPaaS clients to:

- Identify high-value accounts
- Detect low-engagement / potential churn accounts
- Understand hourly traffic patterns for capacity planning and upsell.

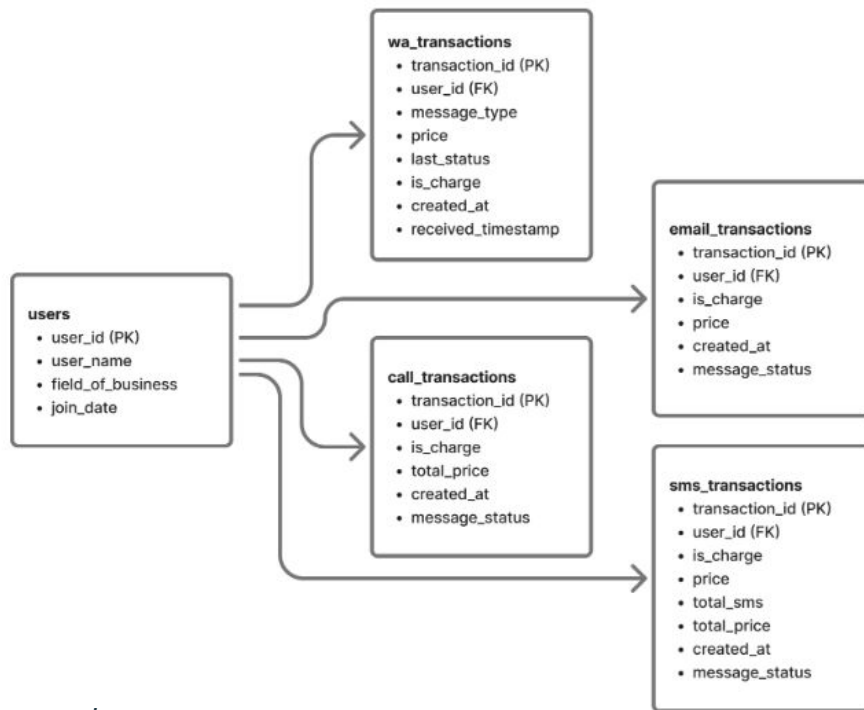
Dataset Scope

Dataset :

- Number of accounts: 20 (90 active day).
- Time scope: 3 months (1 Jan – 31 Mar)
- Tables used:
 - users,
 - wa_transaction,
 - email_transaction,
 - call_transaction,
 - sms_transaction

Disclaimer :

The data provided is a representation of original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company.



Analysis : Google Collab

The Core Metrics Framework

Instead of usual RFM, I mapped the framework to CPaaS Infrastructure Mechanics:

- Recency: cut off in 31 March
- Frequency: Average weekly message volume per account, computed from total volume over weeks in the 90-day .
- Monetary: Total revenue generated by the account.

Methodology & The Percentile Choice

- Data user : 20
- All accounts have the same active period and similar recency
- This makes classic clustering (e.g., K-Means) unstable and hard to interpret with executives for such small n

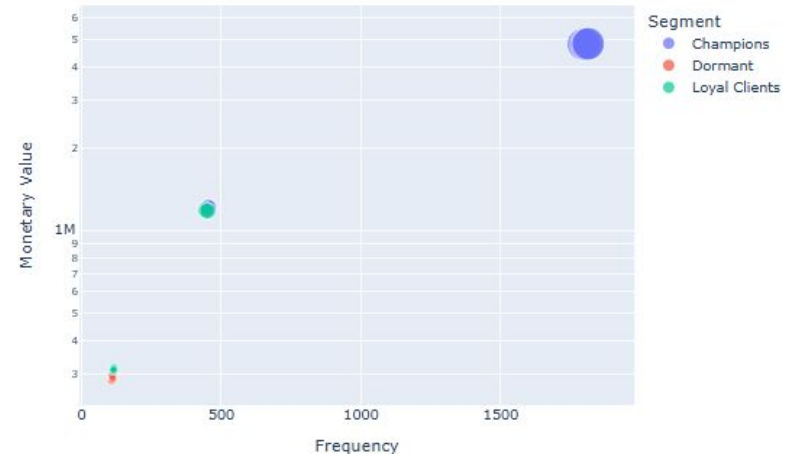
Instead, we use rule-based RFM segmentation with percentiles, which:

- Ranks accounts by Frequency and Monetary
- Assigns percentile-based scores (1–3)
- Creates intuitive segments (Champions, Loyal, Low-Intensity).

User Persona

Champion (R=3, F=3, M=3)	Needs stability, high deliverability, volume discounts, account management.
Loyal Clients (mid-high F & M)	Needs upsell opportunities
Dormant / Low-Intensity (low F, low M)	Need activation campaigns, simple pricing

RFM Scatter Plot: Recency vs Monetary by Segment (Size=Frequency)



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Insight and Recommendation

Segmentation	Revenue	Volume	Problem	Action	Impact
Champions (6 Users)	~Rp 3M	~1,131 msg	11% failure rate during business hours. Failures are not caused by traffic spikes.	During business peak hours automatically route failed high-latency Voice requests into SMS or WhatsApp to decrease fallbacks.	Safeguards the high-volume revenue pipeline without cutting traffic.
Loyal Clients (7 Users)	~Rp 563K	~210 msg	Lowest failure rate (1.23%). Capped at low traffic tiers.	Bundle discount credits to encourage them to increase their frequency in our lower-cost channels.	Grow revenue from already healthy customers.
Low-Intensity (7 Users)	~Rp 295K	~111 msg	High overall failure rate (1.27%). Low financial return.	Migrate to automated self-service with developer playbooks.	Reduce team operational capacity and cuts overhead margins.



Appendix

Python Code

```
df['created_at'] = pd.to_datetime(df['created_at'])
df = df[
    df['created_at'] <= '2025-03-31 23:59:59' ].copy()
df['week'] = df['created_at'].dt.isocalendar().week

snapshot_date = df['created_at'].max() + pd.Timedelta(days=1)

recency = (df.groupby('user_id')['created_at'].max().reset_index())
recency['recency'] = (snapshot_date - recency['created_at']).dt.days
recency = recency[['user_id','recency']]

weekly_volume = (df.groupby(['user_id','week'])['volume'].sum()
    .reset_index(name='weekly_volume'))

frequency = (weekly_volume.groupby('user_id')['weekly_volume']
    .mean().reset_index()
    .rename(columns={'weekly_volume':'frequency'}))

monetary = (df[df['is_charge'] == True].groupby('user_id')['price']
    .sum() .reset_index().rename(columns={'price':'monetary'}))
```

```
rfm = recency.merge(frequency, on='user_id')

rfm = rfm.merge(monetary, on='user_id')

if rfm['recency'].nunique() == 1: rfm['R'] = 3
else:
    rfm['R'] = pd.qcut( rfm['recency'], q=3, labels=[3,2,1], duplicates='drop')

rfm['F'] = pd.qcut( rfm['frequency'].rank(method='first'), q=3, labels=[1,2,3])

rfm['M'] = pd.qcut(rfm['monetary'].rank(method='first'), q=3, labels=[1,2,3])

rfm['RFM_Score'] = (rfm['R'].astype(str) + rfm['F'].astype(str) + rfm['M'].astype(str))

def assign_segment(row):
    if row['R'] == 3 and row['F'] == 3 and row['M'] == 3: return 'Champions'

    elif row['F'] >= 2 and row['M'] >= 2: return 'Loyal Clients'

    elif row['R'] == 1: return 'At Risk'

    Else: return 'Dormant'

rfm['segment'] = rfm.apply( assign_segment, axis=1 )
```



Thank You

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